Gradient-Space Partitioning for Conflict-Reduced Multi-Task Optimization

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Abstract

We propose a geometry-driven gradient-space partitioning objective for conflict-reduced multi-task optimization. We instantiate this framework within a Mixture-of-Experts (MoE) routing architecture as a practical realization. Our method encourages micro-batches with coherent gradient directions to be routed to the same expert, reducing intra-expert gradient conflict while preserving scalable training. We derive a strictly first-order detach implementation and show that a load-normalized alignment objective is equivalent to a relaxed k-means clustering problem in gradient space. We present an experimental plan spanning small-scale controlled setups and 7B QLoRA verification.

1 Introduction

Multi-task optimization often suffers from gradient interference across heterogeneous tasks. When gradients from distinct objectives conflict, their aggregation induces oscillation, noise amplification, and drift in the training dynamics, ultimately slowing convergence and degrading stability.

We propose to view multi-task learning through the lens of gradient-space geometry. Each micro-batch gradient defines a local direction in parameter space, and the optimization trajectory evolves under the aggregate gradient field. When heterogeneous gradients are indiscriminately mixed, the resulting dynamics behave as a superposition of incompatible flows. From this perspective, multi-task optimization can be interpreted as a dynamical system evolving over a union of discontinuous gradient manifolds.

Our key hypothesis is that conflict can be reduced by learning to partition gradient space. By separating micro-batches into geometrically coherent subspaces, each subspace induces a locally consistent optimization flow. Rather than enforcing uniform

load balancing or heuristic task separation, we introduce a geometry-driven control signal that encourages intra-group gradient alignment while discouraging destructive interference.

Sparse expert architectures, such as Mixture-of-Experts (MoE), provide a natural substrate for implementing such partitioning. However, the principle itself is architecture-agnostic: routing becomes a learnable geometric controller that minimizes cross-manifold conflict and stabilizes optimization dynamics.

Our contributions are summarized as follows:

- We reinterpret multi-task optimization as gradient-space partitioning over discontinuous manifolds.
- We introduce a gradient-alignment regularizer that reduces intra-group conflict.
- We propose a load-normalized formulation that preserves specialization without collapse.
- We provide a strictly first-order implementation without second-order overhead.
- We show that the objective is theoretically equivalent to relaxed k-means clustering in gradient space.

2 Background

Consider micro-batches $\{\mathcal{B}_m\}_{m=1}^{M}$ sampled proportionally across tasks. Each micro-batch induces a gradient vector $g_m \in \mathbb{R}^d$ with respect to trainable parameters.

In conventional multi-task optimization systems, the router outputs probabilities

$$p_{mk}, \quad \sum_{k=1}^{K} p_{mk} = 1,$$

and dispatch decisions are made independently of gradient geometry.

However, from a dynamical systems perspective, the training trajectory is governed by the aggregate gradient field. If gradients from heterogeneous tasks exhibit conflicting directions, their mixture induces optimization noise and instability.

We therefore seek to regulate routing using intrinsic geometric signals: experts should receive micro-batches whose gradients are internally coherent, thereby forming locally consistent optimization manifolds.

3 Related Work

Multi-task gradient conflict and gradient surgery. A substantial body of work studies optimization interference in multi-task learning, where gradients from different objectives may point in competing directions, leading to slow convergence and negative transfer. PCGrad [1] performs gradient surgery by projecting conflicting task gradients

to remove negatively aligned components. GradNorm [2] instead balances task learning dynamics by adaptively reweighting losses based on gradient magnitudes. From a multi-objective perspective, MGDA [3] formulates multi-task learning as finding a common descent direction across tasks. These approaches reshape a single global update direction. In contrast, our method does not modify the shared update directly; instead, it partitions gradient space at the micro-batch level, allowing locally coherent optimization flows to emerge through routing.

Token-choice routing and load imbalance in MoE. Sparse Mixture-of-Experts (MoE) architectures typically adopt token-choice routing, where each token selects top-k experts according to router scores. A well-known failure mode is *load imbalance* (router collapse), in which a small subset of experts receives most tokens, leaving others undertrained and wasting capacity. Switch Transformers [4] and DeepSpeed-MoE [5] introduce auxiliary load-balancing losses to mitigate this issue. However, such balancing losses regulate token counts rather than optimization geometry, and they do not explicitly address gradient conflict within experts.

Optimal-transport and assignment-based routing. Some works formulate token-to-expert assignment as a constrained optimization problem, using Sinkhorn normalization or assignment-style routing to approximate balanced dispatch. Routing Transformers [6] and related balancing strategies modify routing semantics from unconstrained expert selection to near-doubly-stochastic assignment. While effective at improving utilization, these methods primarily regulate load distribution and may introduce tension between balancing constraints and specialization dynamics.

Expert-choice routing. Expert-choice routing [7] reverses the assignment direction by allowing experts to select tokens. This alleviates load imbalance while preserving specialization. Nevertheless, routing decisions remain representation-driven rather than optimization-driven.

Our position: routing as geometric control. We reinterpret routing as a geometric control mechanism for conflict-reduced multi-task optimization. Instead of reshaping a single global gradient (as in gradient surgery) or enforcing token-level balancing (as in load-balancing or OT approaches), we partition gradient space into locally coherent submanifolds. Our load-normalized alignment objective encourages intra-expert gradient consistency while discouraging destructive interference. The strictly first-order detach implementation ensures that alignment signals update router parameters only, maintaining scalability.

4 Gradient-Space Partitioning Objective

For each micro-batch m, let $g_m \in \mathbb{R}^d$ denote the gradient of the task loss with respect to the trainable parameters (e.g., LoRA / expert parameters). The router outputs soft probabilities p_{mk} with $\sum_{k=1}^{K} p_{mk} = 1$.

We use a strictly first-order surrogate by detaching the micro-batch gradients:

$$\tilde{g}_m = \mathrm{stopgrad}(g_m).$$

We define the (soft) aggregated gradient for expert k:

$$G_k = \sum_{m=1}^{M} p_{mk}\, \tilde{g}_m, \qquad \mathrm{load}_k = \sum_{m=1}^{M} p_{mk}.$$

Our load-normalized gradient-alignment regularizer is:

$$\mathcal{L}_{\text{norm}} = -\sum_{k=1}^{K} \frac{\|G_k\|^2}{\mathrm{load}_k + \epsilon},$$

where $\epsilon > 0$ is a small constant for numerical stability.

The overall training objective is:

$$\mathcal{L} = \sum_{m=1}^{M} \mathcal{L}_{\text{task}}^{(m)} + \lambda \mathcal{L}_{\text{norm}}.$$

5 Methodology

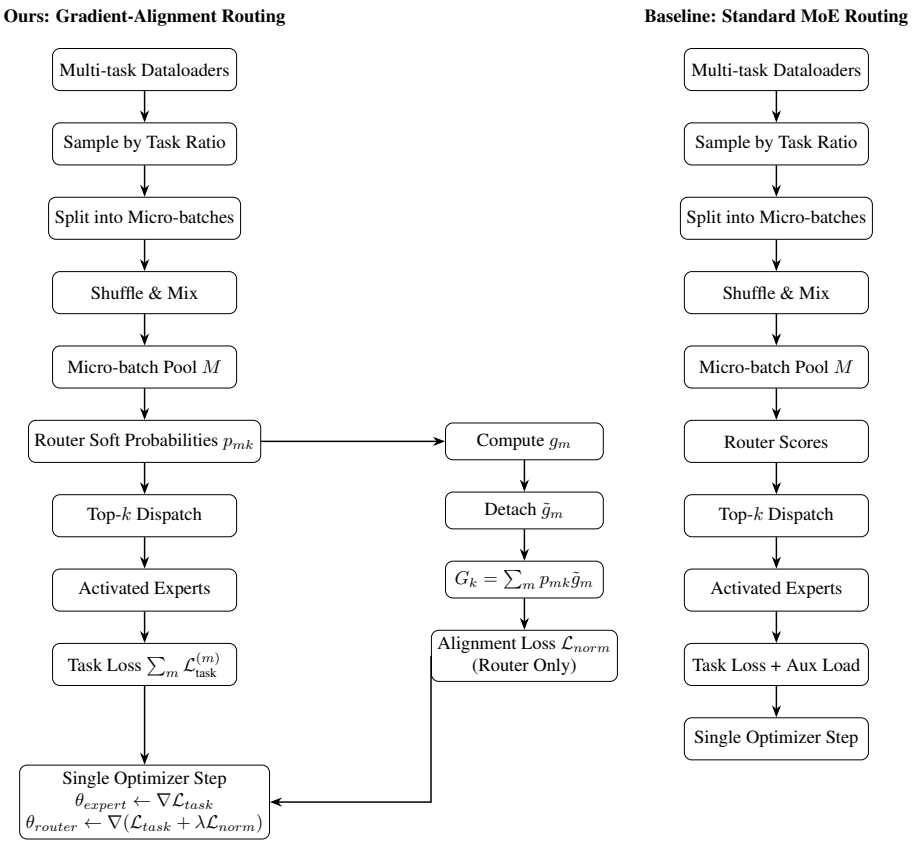

Figure 1: Comparison between standard MoE routing (right) and our gradient-alignment regularized routing (left). Our method introduces a detached gradient aggregation branch that updates router parameters only, while expert parameters remain purely task-driven.

5.1 Micro-Batch Gradient Geometry

Micro-batches represent local regions in data manifold space. Their gradients encode optimization directions. Routing thus becomes a partition of gradient space.

5.2 Alignment Regularizer

Expanding:

$$\|G_k\|^2 = \sum_{i,j} p_{ik} p_{jk} \langle \tilde{g}_i, \tilde{g}_j \rangle .$$

Thus maximizing alignment equals maximizing pairwise inner products within each expert.

5.3 Load-Normalized Variant

Define expert load:

$$\text{load}_k = \sum_m p_{mk}.$$

Normalized objective:

$$\mathcal{L}_{norm} = -\sum_k \frac{\|G_k\|^2}{\text{load}_k}.$$

This emphasizes average intra-expert coherence rather than total magnitude.

5.4 Gradient Dynamics and Credit Assignment Interpretation

We analyze how the load-normalized alignment regularizer reshapes routing probabilities. Recall that

$$G_k = \sum_i p_{ik} \tilde{g}_i, \qquad \mathcal{L}_{\text{norm}} = -\sum_k \frac{\|G_k\|^2}{\text{load}_k}.$$

To understand the mechanism, we first consider the derivative of the squared norm with respect to a routing probability p_{mk}.

Since

$$\|G_k\|^2 = G_k^\top G_k,$$

and

$$\frac{\partial}{\partial G_k} \|G_k\|^2 = 2G_k,$$

by the chain rule we obtain

$$\frac{\partial}{\partial p_{mk}} \|G_k\|^2 = \left(\frac{\partial \|G_k\|^2}{\partial G_k} \right)^\top \frac{\partial G_k}{\partial p_{mk}}.$$

Because

$$\frac{\partial G_k}{\partial p_{mk}} = \tilde{g}_m,$$

we obtain

$$\frac{\partial}{\partial p_{mk}} \|G_k\|^2 = 2\langle G_k, \tilde{g}_m \rangle.$$

Geometric interpretation. The inner product $\langle G_k, \tilde{g}_m \rangle$ measures the alignment between the current aggregated expert direction G_k and the micro-batch gradient $\tilde{g}_m$.

Intuitively:

- If $\tilde{g}_m$ is aligned with G_k, then increasing p_{mk} increases $\|G_k\|^2$.
- If $\tilde{g}_m$ conflicts with G_k, then increasing p_{mk} decreases $\|G_k\|^2$.

Therefore, the alignment regularizer induces a *continuous credit assignment mechanism*: each micro-batch adjusts its routing probability toward experts whose aggregated direction is geometrically consistent with its own gradient, and away from experts with conflicting directions.

Because $\tilde{g}_m$ is detached, this adjustment affects only router parameters. Expert parameters are still updated exclusively by the task loss.

5.5 First-Order Detach Implementation

Without detach, gradients of $\mathcal{L}_{align}$ would involve Hessian terms. By using $\tilde{g}_m = \mathrm{stopgrad}(g_m)$, alignment gradients flow only into router parameters.

Computational complexity per step:

$$O(Md).$$

5.6 Overall Training Procedure

We now summarize the complete forward and backward execution of the proposed method. The key principle is a separation of optimization channels:

- The task loss updates activated expert parameters and the router.
- The alignment regularizer updates the router only.

Forward Pass. Given a multi-task batch, we first split it into micro-batches $\{\mathcal{B}_m\}_{m=1}^{M}$. The router produces soft routing probabilities p_{mk}. Experts are activated according to the routing policy (e.g., top-k or soft mixture), and the task loss $\mathcal{L}_{\text{task}}$ is computed.

Backward Pass (Task Channel). Backpropagation of $\mathcal{L}_{\text{task}}$ updates:

- Activated expert parameters,
- Router parameters.

At this stage we obtain the micro-batch gradients g_m with respect to trainable expert parameters (e.g., LoRA weights).

Backward Pass (Alignment Channel). We detach the micro-batch gradients:

$$\tilde{g}_m = \mathrm{stopgrad}(g_m),$$

and compute the aggregated expert directions:

$$G_k = \sum_m p_{mk} \tilde{g}_m.$$

The alignment loss is then:

$$\mathcal{L}_{\text{norm}} = -\sum_k \frac{\|G_k\|^2}{\text{load}_k + \epsilon}.$$

Backpropagation of $\lambda \mathcal{L}_{\text{norm}}$ updates *router parameters only*, since $\tilde{g}_m$ is detached and does not propagate gradients into experts.

Parameter Update. The final parameter update per iteration is:

- Expert parameters:
$$\theta_{\text{expert}} \leftarrow \theta_{\text{expert}} - \eta \nabla_{\theta_{\text{expert}}} \mathcal{L}_{\text{task}}.$$
- Router parameters:
$$\theta_{\text{router}} \leftarrow \theta_{\text{router}} - \eta \nabla_{\theta_{\text{router}}} \big(\mathcal{L}_{\text{task}} + \lambda \mathcal{L}_{\text{norm}} \big).$$

Computational Complexity. The additional overhead of the alignment channel is $O(Md)$, where M is the number of micro-batches and d the gradient dimension. No second-order derivatives or double backward operations are required.

5.7 Algorithmic Implementation

We summarize the forward and backward execution of our method in Algorithm 1.

Algorithm 1 Gradient-Space Partitioning Training Procedure

Sample multi-task batch $\{\mathcal{B}_m\}_{m=1}^{M}$
Split into micro-batches
Compute router probabilities p_{mk}
Forward pass through activated experts (top-k or soft routing)
Compute task loss $\mathcal{L}_{task}$
Backpropagate $\mathcal{L}_{task}$
Obtain micro-batch gradients g_m
Detach gradients $\tilde{g}_m = \mathrm{stopgrad}(g_m)$
Compute aggregated expert gradients:

$$G_k = \sum_m p_{mk} \tilde{g}_m$$

Compute alignment loss:

$$\mathcal{L}_{norm} = -\sum_k \frac{\|G_k\|^2}{\mathrm{load}_k}$$

Backpropagate $\lambda \mathcal{L}_{norm}$
Update parameters:

- Expert parameters $\leftarrow$ task loss only
- Router parameters $\leftarrow$ task loss + alignment loss

5.8 Equivalence to Gradient-Space k-Means and Spectral Relaxation

Theorem 1 (Equivalence to k-means under hard routing). Under hard routing, minimizing the load-normalized alignment loss is equivalent to minimizing the classical k-means objective in gradient space.

Proof. Let $\{g_m\}_{m=1}^{M} \subset \mathbb{R}^d$ be micro-batch gradients and assume hard routing

$$p_{mk} \in \{0, 1\}, \quad \mathcal{C}_k = \{m \mid p_{mk} = 1\}.$$

Let $|\mathcal{C}_k|$ denote cluster size. Classical k-means minimizes

$$\mathcal{J}_{\text{k-means}} = \sum_{k=1}^{K} \sum_{m \in \mathcal{C}_k} \|g_m - \mu_k\|^2, \qquad \mu_k = \frac{1}{|\mathcal{C}_k|} \sum_{m \in \mathcal{C}_k} g_m.$$

Expanding,

$$\|g_m - \mu_k\|^2 = \|g_m\|^2 - 2\langle g_m, \mu_k \rangle + \|\mu_k\|^2.$$

Summing over $m \in \mathcal{C}_k$ and substituting μ_k yields the identity

$$\sum_{m \in \mathcal{C}_k} \|g_m - \mu_k\|^2 = \sum_{m \in \mathcal{C}_k} \|g_m\|^2 - \frac{1}{|\mathcal{C}_k|} \left\| \sum_{m \in \mathcal{C}_k} g_m \right\|^2 .$$

The first term $\sum_{m=1}^{M} \|g_m\|^2$ is independent of the clustering; therefore minimizing $\mathcal{J}_{k\text{-means}}$ is equivalent to maximizing

$$\sum_{k=1}^{K} \frac{1}{|\mathcal{C}_k|} \left\| \sum_{m \in \mathcal{C}_k} g_m \right\|^2 .$$

Under hard routing, $G_k = \sum_{m \in \mathcal{C}_k} g_m$ and $\text{load}_k = |\mathcal{C}_k|$, hence our loss

$$\mathcal{L}_{norm} = -\sum_{k=1}^{K} \frac{\|G_k\|^2}{\text{load}_k} = -\sum_{k=1}^{K} \frac{1}{|\mathcal{C}_k|} \left\| \sum_{m \in \mathcal{C}_k} g_m \right\|^2$$

is exactly the negative of the k-means maximization form above, proving

$$\arg\min \mathcal{J}_{k\text{-means}} \iff \arg\min \mathcal{L}_{norm}.$$

□

Interpretation. Experts correspond to clusters in gradient space, and routing corresponds to assignments. Soft routing probabilities $p_{mk} \in [0, 1]$ therefore induce a relaxed (soft) k-means formulation.

Theorem 2 (Spectral relaxation of load-normalized alignment; Ratio Association). Under hard routing, we have already shown in Theorem 1 that

$$\text{load}_k = |\mathcal{C}_k|, \qquad G_k = \sum_{m \in \mathcal{C}_k} g_m.$$

Therefore the load-normalized alignment loss becomes

$$\mathcal{L}_{norm} = -\sum_{k=1}^{K} \frac{1}{|\mathcal{C}_k|} \left\| \sum_{m \in \mathcal{C}_k} g_m \right\|^2 .$$

Let $\{g_m\}_{m=1}^{M} \subset \mathbb{R}^d$ and define the gradient affinity matrix

$$W \in \mathbb{R}^{M \times M}, \qquad W_{ij} := \langle g_i, g_j \rangle .$$

Then minimizing $\mathcal{L}_{norm}$ is equivalent to maximizing the *ratio association* objective

$$\max_{\{\mathcal{C}_k\}_{k=1}^{K}} \sum_{k=1}^{K} \frac{1}{|\mathcal{C}_k|} \sum_{i,j \in \mathcal{C}_k} W_{ij}.$$

Moreover, its standard continuous relaxation yields a top-K eigenvector solution.

Proof. First expand each cluster term:

$$\left\| \sum_{m \in \mathcal{C}_k} g_m \right\|^2 = \sum_{i \in \mathcal{C}_k} \sum_{j \in \mathcal{C}_k} \langle g_i, g_j \rangle = \sum_{i,j \in \mathcal{C}_k} W_{ij}.$$

Hence

$$-\mathcal{L}_{norm} = \sum_{k=1}^{K} \frac{1}{|\mathcal{C}_k|} \sum_{i,j \in \mathcal{C}_k} W_{ij},$$

which is exactly the ratio association objective.

To express it in trace form, let $H \in \{0,1\}^{M \times K}$ be the hard assignment matrix, where $H_{ik} = 1$ iff $i \in \mathcal{C}_k$. Define

$$S := H^\top H = \mathrm{diag}(|\mathcal{C}_1|, \ldots, |\mathcal{C}_K|).$$

Let h_k denote the k-th column of H. Then

$$\sum_{i,j \in \mathcal{C}_k} W_{ij} = h_k^\top W h_k, \qquad |\mathcal{C}_k| = h_k^\top h_k.$$

Therefore

$$-\mathcal{L}_{norm} = \sum_{k=1}^{K} \frac{h_k^\top W h_k}{h_k^\top h_k} = \mathrm{Tr}\big(S^{-1} H^\top W H\big)\,.$$

Define the normalized embedding

$$Y := H S^{-1/2} \in \mathbb{R}^{M \times K}, \qquad Y^\top Y = I_K.$$

Then

$$-\mathcal{L}_{norm} = \mathrm{Tr}\big(Y^\top W Y\big)\,,$$

where Y is constrained to arise from a discrete partition.

Relaxing the discreteness constraint and optimizing over all $Y \in \mathbb{R}^{M \times K}$ satisfying $Y^\top Y = I_K$ gives

$$\max_{Y^\top Y = I_K} \mathrm{Tr}(Y^\top W Y).$$

By the Rayleigh–Ritz theorem, the maximizer is given by choosing the columns of Y as the top-K eigenvectors of W. □

Interpretation. The load-normalized alignment term admits a spectral relaxation: it corresponds to ratio association graph partitioning on the gradient affinity graph W. Hard routing recovers a discrete partition, while the continuous relaxation corresponds to a spectral embedding spanned by the top-K eigenvectors of W.

5.9 Dynamical-System Analysis of Routing: Collapse and Anti-Collapse

5.9.1 Idealized Theoretical Assumptions

To rigorously derive the limiting behavior of soft routing, we analyze an idealized symmetric setting. The goal is not to fully model practical training, but to isolate the core geometric and dynamical mechanisms governing collapse and anti-collapse.

Expert–manifold matching ($K = M$). We assume the number of experts equals the number of latent gradient manifolds:

$$K = M.$$

Each manifold $c \in \{1, \ldots, M\}$ is associated with a prototype unit direction $u_c \in \mathbb{R}^d$, $\|u_c\| = 1$. The ideal optimal configuration assigns one expert to each manifold.

Uniform manifold prior. Training micro-batches are drawn from the M manifolds with equal probability:

$$\Pr(c) = \frac{1}{M}.$$

Thus, no manifold has intrinsic frequency advantage. Any imbalance arises purely from routing dynamics.

Gradient concentration model. Detached micro-batch gradients follow

$$\tilde{g} \mid c = u_c + \xi,$$

where

$$\mathbb{E}[\xi \mid c] = 0, \qquad \mathrm{Cov}(\xi \mid c) \preceq \sigma^2 I.$$

Hence,

$$\mathbb{E}[\tilde{g} \mid c] = u_c.$$

Intra- and inter-manifold coherence. We define the expected pairwise cosine similarity:

$$\mu_{\text{intra}} = \mathbb{E}[\langle \tilde{g}_i, \tilde{g}_j \rangle \,|\, c_i = c_j]\,,$$

$$\mu_{\text{inter}} = \mathbb{E}[\langle \tilde{g}_i, \tilde{g}_j \rangle \,|\, c_i \neq c_j]\,.$$

We assume

$$\mu_{\text{intra}} > 0, \qquad \mu_{\text{inter}} < \mu_{\text{intra}}.$$

In particular, the anti-collapse regime corresponds to

$$\mu_{\text{inter}} < 0,$$

i.e., gradients from different manifolds exhibit destructive interference.

Load and mixture within an expert. Let L_k denote the total routed load to expert k within a short analysis window. Let π_{kc} denote the fraction of mode-c samples within expert k, with $\sum_{c=1}^{M} \pi_{kc} = 1$.

Under the concentration model, the aggregated direction satisfies

$$\mathbb{E}\|G_k\|^2 = L_k^2 \left(\sum_c \pi_{kc}^2 \mu_{\text{intra}} + \sum_{c \neq c'} \pi_{kc} \pi_{kc'} \mu_{\text{inter}} \right). \tag{1}$$

Equivalently,

$$\mathbb{E}\|G_k\|^2 = L_k^2 \left(\mu_{\text{inter}} + (\mu_{\text{intra}} - \mu_{\text{inter}}) \sum_c \pi_{kc}^2 \right). \tag{2}$$

Geometric implication of mixing. Since $\sum_c \pi_{kc}^2$ is maximized (equal to 1) when expert k receives a single manifold, and minimized (equal to $1/M$) under uniform mixing, Equation (2) shows:

- Pure (single-manifold) experts maximize $\|G_k\|^2$.
- Mixed experts suffer coherence reduction proportional to $|\mu_{\text{inter}}|$.
- The stronger the conflict ($|\mu_{\text{inter}}|$ large), the larger the geometric penalty of mixing.

Initial symmetry. We assume symmetric initialization:

$$p_k(0) = \frac{1}{K}, \qquad s_k(0) = s_0.$$

Thus, no expert has intrinsic advantage. Any asymmetry emerges dynamically.

Continuous-time routing abstraction. We analyze routing via a continuous-time approximation of soft probability updates, treating

$$p(t) \in \Delta^{K-1}.$$

This allows direct stability and limit analysis.

5.9.2 Baseline Dynamics: From Small Asymmetry to Vertex Collapse

Under the idealized assumptions in §5.9.1, we first analyze the baseline case where routing is driven solely by task loss (no alignment regularizer). We show that even under a symmetric initialization, an arbitrarily small early advantage can be amplified into a winner-take-all collapse (a simplex vertex attractor) via a positive feedback loop between load allocation and learning speed.

(1) Uniform start with a small perturbation. We consider a soft routing state $p(t) \in \Delta^{K-1}$ with symmetric initialization $p_k(0) = 1/K$. A "small perturbation" means that for some pair of experts $i \neq j$,

$$p_i(0) = \frac{1}{K} + \varepsilon, \qquad p_j(0) = \frac{1}{K} - \varepsilon, \qquad 0 < \varepsilon \ll 1,$$

and optionally a matching competence perturbation $s_i(0) > s_j(0)$. Such asymmetry can arise from stochastic optimization noise or finite-sample effects.

Competence–loss coupling and load-driven learning. As introduced in the baseline collapse analysis, we assume

$$\ell_k(t) = \ell_0 - a\, s_k(t), \qquad a > 0, \tag{3}$$

and load-driven competence dynamics

$$\dot{s}_k(t) = \alpha\, p_k(t), \qquad \alpha > 0. \tag{4}$$

(2) Ratio dynamics $\Rightarrow$ positive feedback. We model the baseline router update by replicator dynamics with payoff $\Pi_k(t) = -\ell_k(t)$:

$$\dot{p}_k(t) = \beta\, p_k(t)\big(\bar{\ell}(t) - \ell_k(t)\big), \qquad \beta > 0, \tag{5}$$

where $\bar{\ell}(t) = \sum_{j=1}^{K} p_j(t)\ell_j(t)$. For any two experts i, j, define $r_{ij}(t) = p_i(t)/p_j(t)$. A standard identity of replicator dynamics gives

$$\frac{d}{dt} \log r_{ij}(t) = \beta\big(\ell_j(t) - \ell_i(t)\big). \tag{6}$$

Substituting (3) yields

$$\frac{d}{dt} \log r_{ij}(t) = \beta a\big(s_i(t) - s_j(t)\big). \tag{7}$$

Meanwhile, from (4),

$$\frac{d}{dt}\big(s_i(t) - s_j(t)\big) = \alpha\big(p_i(t) - p_j(t)\big). \tag{8}$$

Equations (7)–(8) form a closed positive feedback loop: a small competence advantage increases routing probability, which further increases competence.

(3) Strict collapse in the two-expert case ($K = 2$). The analytically tractable case is $K = 2$. Let $p_2(t) = 1 - p_1(t)$ and define the competence gap $\Delta s(t) = s_1(t) - s_2(t)$. From (8),

$$\dot{\Delta s}(t) = \alpha(2p_1(t) - 1), \tag{9}$$

and from (7) with $r(t) = p_1(t)/(1 - p_1(t))$,

$$\frac{d}{dt} \log \frac{p_1(t)}{1 - p_1(t)} = \beta a\, \Delta s(t). \tag{10}$$

Proposition 5.9.2 (winner-take-all under a small initial advantage). Assume $K = 2$ and there exists an initial advantage such that

$$p_1(0) > \tfrac{1}{2} \quad \text{and} \quad \Delta s(0) > 0.$$

Then $\Delta s(t)$ is strictly increasing and

$$\frac{p_1(t)}{1 - p_1(t)} \to \infty \quad \Rightarrow \quad p_1(t) \to 1.$$

Proof. Since $p_1(0) > \frac{1}{2}$, (9) gives $\dot{\Delta s}(0) > 0$. As long as $p_1(t) > \frac{1}{2}$, we have $\dot{\Delta s}(t) > 0$, hence $\Delta s(t) \geq \Delta s(0) > 0$. Then (10) implies $\frac{d}{dt} \log \frac{p_1}{1-p_1} = \beta a \, \Delta s(t) \geq \beta a \, \Delta s(0) > 0$, so $\log \frac{p_1(t)}{1-p_1(t)}$ grows without bound and therefore $p_1(t) \to 1$. □

Extension to $K > 2$ (pairwise domination). For general K, the ratio identity (6) holds for every pair (i, j). If an expert $k^\star$ maintains a competence advantage $s_{k^\star}(t) > s_j(t)$ over all $j \neq k^\star$ for a non-trivial period, then $\log(p_{k^\star}/p_j)$ increases for all j, driving $p_{k^\star}(t) \to 1$ and $p_j(t) \to 0$. Thus, without an explicit geometric regularizer, soft routing exhibits an intrinsic tendency toward vertex collapse driven by early stochastic asymmetry.

5.9.3 Ours – Case 1: Homogeneous Mixing with Unequal Load

We now analyze the first regime under the alignment-augmented payoff. Assume that each expert receives a statistically homogeneous mixture of manifolds, i.e.,

$$\pi_{kc} = \frac{1}{M} \quad \text{for all } k, c,$$

so that all experts have identical internal conflict statistics, but their total loads L_k may differ.

Expected coherence under homogeneous mixing. Substituting $\pi_{kc} = 1/M$ into (2) yields

$$\mathbb{E}\|G_k\|^2 = L_k^2 \left(\mu_{\text{inter}} + (\mu_{\text{intra}} - \mu_{\text{inter}}) \frac{1}{M} \right). \tag{11}$$

Define the constant

$$\gamma := \mu_{\text{inter}} + (\mu_{\text{intra}} - \mu_{\text{inter}}) \frac{1}{M}.$$

Under the anti-collapse regime $\mu_{\text{inter}} < 0$ and $\mu_{\text{intra}} > 0$, we have $\gamma > 0$ provided intra-manifold coherence dominates.

Thus,

$$\mathbb{E}\|G_k\|^2 = \gamma L_k^2.$$

Alignment utility scaling. Recall that

$$U_k = \frac{\|G_k\|^2}{L_k + \epsilon}.$$

Taking expectation and ignoring ϵ for clarity,

$$\mathbb{E}[U_k] \approx \frac{\gamma L_k^2}{L_k} = \gamma L_k. \tag{12}$$

Hence, alignment utility scales linearly with load.

Marginal alignment gain. However, routing decisions depend on *marginal utility*. From (13),

$$\frac{\partial U_k}{\partial p_{mk}} = \frac{2\langle G_k, \tilde{g}_m \rangle}{L_k} - \frac{\|G_k\|^2}{L_k^2}. \tag{13}$$

Under homogeneous mixing, $\mathbb{E}\langle G_k, \tilde{g}_m \rangle \propto L_k$, and $\mathbb{E}\|G_k\|^2 \propto L_k^2$. Therefore,

$$\mathbb{E}\left[\frac{\partial U_k}{\partial p_{mk}}\right] \propto \frac{L_k}{L_k} - \frac{L_k^2}{L_k^2} = \text{constant} - \text{constant}.$$

More precisely, since both numerator terms scale with L_k, the second (negative) term becomes relatively more dominant for larger L_k, leading to diminishing marginal gain as load increases.

Proposition (Anti-amplification under homogeneous conflict). Under homogeneous mixing ($\pi_{kc} = 1/M$), the marginal alignment utility of an expert is a non-increasing function of its load. Consequently, the alignment term introduces a negative-feedback effect that suppresses further amplification of load imbalance.

Interpretation. In contrast to the baseline dynamics, where higher load strictly increases learning speed and further increases routing probability, the alignment term produces diminishing returns: overloaded experts do not gain proportionally larger marginal payoff. Thus, even without geometric specialization, the alignment regularizer attenuates winner-take-all amplification.

5.9.4 Ours – Case 2: Infinitesimal Bias ⇒ Directional Tilt ⇒ Geometric Positive Feedback

We now analyze the second regime: even an arbitrarily small routing bias toward a manifold induces a directional tilt of the expert aggregate gradient, which in turn increases the alignment utility for same-manifold samples and decreases it for conflicting manifolds. This creates a *geometric positive feedback* that drives specialization.

Setup (mode prototype model). Under the idealized setting in §5.9.1, micro-batch detached gradients satisfy

$$\mathbb{E}[\tilde{g} \mid c] = u_c, \qquad \|u_c\| = 1,$$

with intra- and inter-manifold coherence parameters

$$\mu_{\text{intra}} = \mathbb{E}[\langle \tilde{g}_i, \tilde{g}_j \rangle \,|\, c_i = c_j]\,, \qquad \mu_{\text{inter}} = \mathbb{E}[\langle \tilde{g}_i, \tilde{g}_j \rangle \,|\, c_i \neq c_j]\,,$$

where $\mu_{\text{inter}} < \mu_{\text{intra}}$, and in the conflict regime $\mu_{\text{inter}} < 0$.

Let expert k receive a mixture of manifolds with proportions π_{kc} ($\sum_c \pi_{kc} = 1$) and total load L_k over a short window.

Lemma 1 (Directional tilt of aggregated gradient). Conditioned on the mixture proportions $\pi_{k\cdot}$, the aggregated detached direction

$$G_k = \sum_{m=1}^{M} p_{mk}\, \tilde{g}_m$$

satisfies

$$\mathbb{E}[G_k \mid \pi_{k\cdot}] = L_k \sum_{c=1}^{M} \pi_{kc}\, u_c. \tag{14}$$

Proof. Let $\mathcal{I}_k$ be the set of micro-batches routed to expert k within the window, so $|\mathcal{I}_k| = L_k$ and the empirical fraction of mode c equals π_{kc}. Then

$$\mathbb{E}[G_k \mid \pi_{k\cdot}] = \sum_{m \in \mathcal{I}_k} \mathbb{E}[\tilde{g}_m \mid c_m] = \sum_{c=1}^{M} \big(L_k \pi_{kc}\big)\, u_c = L_k \sum_c \pi_{kc}\, u_c. \square$$

Interpretation. Equation (14) shows that any small bias in π_{kc} immediately tilts the mean direction of G_k toward the corresponding prototype u_c.

Lemma 2 (Tilt implies same-manifold alignment advantage). For a sample from manifold c, the expected alignment with expert k satisfies

$$\mathbb{E}[\langle \mathbb{E}[G_k \mid \pi_{k\cdot}],\, \tilde{g} \rangle \,|\, c] = L_k \left(\pi_{kc}\, \mu_{\text{intra}} + \sum_{c' \neq c} \pi_{kc'}\, \mu_{\text{inter}} \right). \tag{15}$$

In particular, holding $\{\pi_{kc'}\}_{c' \neq c}$ fixed, the right-hand side is strictly increasing in π_{kc} whenever $\mu_{\text{intra}} > \mu_{\text{inter}}$.

Proof. Using Lemma 5.9.4, conditioning on c and linearity,

$$\mathbb{E}[\langle \mathbb{E}[G_k \mid \pi],\, \tilde{g} \rangle \mid c] = L_k \sum_{c'} \pi_{kc'}\, \mathbb{E}[\langle u_{c'}, \tilde{g} \rangle \mid c]\,.$$

By definition of $\mu_{\text{intra}}, \mu_{\text{inter}}$, we have $\mathbb{E}[\langle u_c, \tilde{g} \rangle \mid c] = \mu_{\text{intra}}$ and for $c' \neq c$, $\mathbb{E}[\langle u_{c'}, \tilde{g} \rangle \mid c] = \mu_{\text{inter}}$, yielding (15). $\square$

Interpretation. Once an expert k is slightly enriched in manifold c (larger π_{kc}), samples from manifold c become *more aligned* with G_k in expectation, while samples from other manifolds become less aligned when $\mu_{\text{inter}} < \mu_{\text{intra}}$ (and especially when $\mu_{\text{inter}} < 0$).

Lemma 3 (Tilt increases per-load coherence utility for purer mixtures). Recall the expected coherence formula (2):

$$\mathbb{E}\|G_k\|^2 = L_k^2\Big(\mu_{\text{inter}} + (\mu_{\text{intra}} - \mu_{\text{inter}})\sum_c \pi_{kc}^2\Big).$$

Thus, for fixed L_k, the expected utility

$$\mathbb{E}[U_k] \approx \frac{\mathbb{E}\|G_k\|^2}{L_k} = L_k\Big(\mu_{\text{inter}} + (\mu_{\text{intra}} - \mu_{\text{inter}})\sum_c \pi_{kc}^2\Big)$$

is strictly increasing in the *purity measure* $\sum_c \pi_{kc}^2$ whenever $\mu_{\text{intra}} > \mu_{\text{inter}}$.

Consequent geometric positive feedback (specialization loop). Consider a manifold $c^\star$ and an expert k with a tiny initial enrichment

$$\pi_{kc^\star} = \frac{1}{M} + \delta, \qquad \delta > 0 \text{ arbitrarily small},$$

with the remaining mass spread across other manifolds. Then:

- By Lemma 5.9.4, G_k tilts toward $u_{c^\star}$.
- By Lemma 5.9.4, mode-$c^\star$ samples have larger expected alignment $\mathbb{E}\langle G_k, \tilde{g}\rangle$ than under uniform mixing.
- By Lemma 5.9.4, the expert's coherence utility U_k increases as its mixture becomes purer.

Therefore, under the router payoff with alignment

$$\Pi_k = -\ell_k + \lambda U_k,$$

routing updates tend to *reinforce* the enrichment: mode-$c^\star$ samples are increasingly attracted to expert k (higher alignment benefit), while conflicting modes (especially when $\mu_{\text{inter}} < 0$) experience reduced or negative marginal gain, and are pushed toward other experts. This establishes a geometric positive feedback loop that promotes expert specialization.

Remark (Why this is not the baseline collapse). The baseline positive feedback amplifies *load* through competence and loss, which can collapse to a single expert even without geometric structure. In contrast, the loop above amplifies *purity* in gradient space: it selectively attracts same-manifold samples and repels conflicting ones, which is the precursor mechanism required to destabilize vertex collapse in Case 3.

5.9.5 Ours – Case 3: Vertex Collapse is Unstable (Proof by Contradiction)

We now prove that, under the geometric mode assumptions in §5.9.1, a collapse to a single expert is *not* a locally stable equilibrium once the alignment channel is enabled ($\lambda > 0$). The proof does not require a closed-form solution of the ODE; instead, we show that the collapsed vertex is *not even a local maximizer* of the alignment objective. Hence, any router update that (approximately) follows the alignment gradient admits a direction that pushes probability mass away from the vertex.

Alignment surrogate as a purity functional. Recall the load-normalized utility

$$U_k \;=\; \frac{\|G_k\|^2}{L_k + \epsilon}.$$

Under §5.9.1, ignoring ϵ for clarity and taking expectation, (2) implies

$$\mathbb{E}[U_k] \;\approx\; \frac{\mathbb{E}\|G_k\|^2}{L_k} = L_k\Big(\mu_{\text{inter}} + (\mu_{\text{intra}} - \mu_{\text{inter}}) \sum_{c=1}^{M} \pi_{kc}^2\Big). \tag{16}$$

Summing over experts yields the expected alignment reward

$$\mathcal{R}(p) := \sum_{k=1}^{K} \mathbb{E}[U_k] = \mu_{\text{inter}} \sum_k L_k + (\mu_{\text{intra}} - \mu_{\text{inter}}) \sum_k L_k \sum_c \pi_{kc}^2. \tag{17}$$

Since $\sum_k L_k$ is fixed by the total number of routed micro-batches in the analysis window, maximizing $\mathcal{R}$ is equivalent to maximizing the *weighted purity* functional

$$\Phi := \sum_{k=1}^{K} L_k \sum_{c=1}^{M} \pi_{kc}^2. \tag{18}$$

Because $\mu_{\text{intra}} > \mu_{\text{inter}}$ by assumption, increasing Φ strictly increases $\mathcal{R}$.

Collapsed vertex implies maximal mixing. Consider the (collapsed) vertex state in which a single expert $k^\star$ receives all load:

$$L_{k^\star} = L, \qquad L_j = 0 \;\; (j \neq k^\star).$$

Under the uniform manifold prior $\Pr(c) = 1/M$, this implies the dominant expert receives the entire mixture, i.e.,

$$\pi_{k^\star c} = \frac{1}{M} \quad \forall c.$$

Then the weighted purity equals

$$\Phi_{\text{coll}} = L \sum_{c=1}^{M} \left(\frac{1}{M}\right)^2 = \frac{L}{M}. \tag{19}$$

A separating perturbation strictly increases purity. We now show that the collapsed state is not locally optimal for Φ by constructing an arbitrarily small perturbation that increases it.

Fix any manifold c_0. Move an infinitesimal amount of load $\delta > 0$ consisting solely of mode-c_0 micro-batches from expert $k^\star$ to an unused expert j. Then

$$L_j = \delta, \qquad L_{k^\star} = L - \delta.$$

Expert j becomes pure for c_0, hence

$$\pi_{jc_0} = 1, \qquad \pi_{jc} = 0 \ (c \neq c_0), \quad \Rightarrow \quad \sum_c \pi_{jc}^2 = 1.$$

For the remaining dominant expert $k^\star$, the mixture becomes *slightly less uniform*; a first-order expansion (in δ/L) gives

$$\sum_c \pi_{k^\star c}^2 = \frac{1}{M} + O\left(\frac{\delta}{L}\right). \tag{20}$$

Therefore, the perturbed weighted purity satisfies

$$\begin{aligned} \Phi_{\text{pert}} &= (L-\delta)\sum_c \pi_{k^\star c}^2 \ + \ \delta \sum_c \pi_{jc}^2 \\ &= (L-\delta)\left(\frac{1}{M} + O\left(\frac{\delta}{L}\right)\right) \ + \ \delta \cdot 1 \\ &= \frac{L}{M} \ + \ \delta\left(1 - \frac{1}{M}\right) \ + \ O\left(\frac{\delta^2}{L}\right). \end{aligned} \tag{21}$$

For any $M > 1$, the first-order term is strictly positive:

$$\Phi_{\text{pert}} - \Phi_{\text{coll}} = \delta\left(1 - \frac{1}{M}\right) + O\left(\frac{\delta^2}{L}\right) > 0 \quad \text{for sufficiently small } \delta.$$

Hence Φ (and thus $\mathcal{R}$) can be strictly increased by an arbitrarily small separating perturbation.

Theorem (vertex instability under alignment-driven routing). Assume the idealized setting in §5.9.1 with $M > 1$ and $\mu_{\text{intra}} > \mu_{\text{inter}}$. Consider router dynamics that includes an alignment channel with $\lambda > 0$ and that, locally, performs ascent on the alignment reward $\mathcal{R}(p)$ (e.g., via a gradient flow approximation or a replicator-style update with payoff containing U_k). Then any collapsed vertex state (all load on a single expert) is not locally asymptotically stable. Equivalently, there exists an arbitrarily small perturbation of routing that strictly increases $\mathcal{R}$, so the alignment-induced update admits an outward direction from the vertex.

Proof (by contradiction). If a collapsed vertex were locally asymptotically stable under the alignment channel, it would have to be a local maximizer of $\mathcal{R}$ (or at least admit no improving local direction) in a neighborhood. However, the construction above produces, for any neighborhood (choose δ small enough), a perturbation that increases Φ and thus increases $\mathcal{R}$ because $\mu_{\text{intra}} - \mu_{\text{inter}} > 0$ in (17). This contradicts local maximality, hence the vertex cannot be locally asymptotically stable. □

Discussion (why this prevents collapse). Baseline collapse arises from a load–learning positive feedback: more load $\Rightarrow$ faster learning $\Rightarrow$ lower loss $\Rightarrow$ more load. The alignment channel introduces a competing geometric pressure: mixed experts reduce the weighted purity Φ and thereby reduce the attainable alignment reward. Any infinitesimal separation of modes across experts increases Φ (by convexity of squares), creating an outward direction from simplex vertices. Thus, once $\lambda > 0$, collapse-to-a-single-expert is dynamically disfavored.

Remark (interaction with task loss). The result above isolates the alignment channel. In the full payoff $\Pi_k = -\ell_k + \lambda U_k$, the task-loss term may still prefer a dominant expert if its competence is significantly higher. Case 1 and Case 2 explain how the alignment channel counteracts runaway amplification and promotes geometric specialization, so that the system can move toward multi-expert utilization rather than locking into a single vertex.

6 Experimental Plan

6.1 Stage A: Controlled Small Models

Goal: Verify conflict reduction.

Metrics:

- Average cosine similarity within experts
- Training stability
- Validation accuracy

Ablations:

- No alignment
- Alignment without normalization
- Normalized variant
- No detach (second-order baseline)

6.2 Stage B: 7B QLoRA Verification

Goal: Validate scalability.

Setup:

- 4-bit base model
- LoRA trainable parameters
- Gradients computed only on LoRA

Metrics:

- Instruction-following benchmarks
- Gradient norm variance
- Routing entropy

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